

A ZFC Continuum-Sized Almost Disjoint Family on $\mathbb{N}^*$

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Status

Candidate partial result, likely valid. The packet gives a ZFC lower bound for Question 4.3 of Lambie-Hanson–Schrittesser [1]. It does not settle the full requested cardinalities $2^{\aleph_1}$ or $2^{2^{\aleph_0}}$ in all models.

Source Question

The source paper defines an almost disjoint family on a topological space X to be a family of noncompact regular open subsets of X whose pairwise intersections are compact. Question 4.3 asks:

Is it provable in ZFC that there exists an almost disjoint family on $\mathbb{N}^*$ of cardinality $2^{\aleph_1}$, or even of cardinality $2^{2^{\aleph_0}}$?

The source crop from page 10 is:

operator $F : c_0(\omega_1) \rightarrow \ell_\infty([\omega]^\omega)$: in [2, Theorem 1], it is proven that CH implies that $\text{Ext}^2(c_0(\omega_1), c_0) \neq 0$. It is at least conceivable that the opposite answer might follow from an assumption such as the Proper Forcing Axiom.

Finally, we wonder whether an assumption like $\mathfrak{b} = 2^{\aleph_0}$ is necessary to construct a large almost disjoint family on $\mathbb{N}^*$.

Question 4.3. *Is it provable in ZFC that there exists an almost disjoint family on $\mathbb{N}^*$ of cardinality $2^{\aleph_1}$, or even of cardinality $2^{2^{\aleph_0}}$?*

Idea of the Proof

The construction embeds the ordinary almost disjoint-family trick into $\mathbb{N}^*$, but each ordinary almost disjoint set is expanded into a noncompact regular open set. A branch $A_x \subseteq \omega$ is split into infinitely many infinite blocks. The finite initial unions of these blocks give an increasing sequence of clopen subsets of $\mathbb{N}^*$; their union is open and noncompact. Regularity is the only point to check. If an ultrafilter is outside the union, then every basic neighborhood of it can be thinned either outside the branch A_x , or to one point in infinitely many tail blocks, and the resulting basic clopen set misses the union.

Theorem 1. *In ZFC, there is an almost disjoint family on $\mathbb{N}^*$, in the sense of Definition 2.1 of [1], of cardinality $2^{\aleph_0}$.*

Proof. Identify $2^{<\omega}$ with ω . For each branch $x \in 2^\omega$, let

$$A_x = \{x \upharpoonright n : n < \omega\} \subseteq 2^{<\omega} \cong \omega.$$

Then $\{A_x : x \in 2^\omega\}$ is a classical almost disjoint family on ω of size $2^{\aleph_0}$.

For each $x \in 2^\omega$, fix a partition

$$A_x = \bigsqcup_{n < \omega} A_{x,n}$$

into infinite sets, and for $n \geq 1$ set

$$b_{x,n} = \bigcup_{m < n} A_{x,m}.$$

For an infinite $a \subseteq \omega$, write

$$N_a = \{p \in \mathbb{N}^* : a \in p\}$$

for the corresponding basic clopen subset of $\mathbb{N}^*$. Define

$$U_x = \bigcup_{n \geq 1} N_{b_{x,n}}.$$

Each U_x is open.

We first show that U_x is regular open. It is enough to prove that every point outside U_x is outside the interior of $\overline{U_x}$. Fix $p \in \mathbb{N}^* \setminus U_x$, and let N_y be a basic neighborhood of p , so $y \in p$ is infinite. Since $p \notin U_x$, we have $b_{x,n} \notin p$ for every $n \geq 1$. Therefore $y \setminus b_{x,n}$ is infinite for every n ; otherwise $y \subseteq^* b_{x,n}$, which would force $b_{x,n} \in p$.

If $y \setminus A_x$ is infinite, choose an infinite $z \subseteq y \setminus A_x$. Then $z \cap b_{x,n} = \emptyset$ for all n , so $N_z \subseteq N_y$ is disjoint from U_x .

Otherwise $y \setminus A_x$ is finite. Since $y \setminus b_{x,n}$ is infinite for every n , the set y meets infinitely many blocks $A_{x,k}$. Choose an infinite $z \subseteq y \cap A_x$ containing at most one point from each block $A_{x,k}$. Then $z \cap b_{x,n}$ is finite for every n . Hence $N_z \cap N_{b_{x,n}} = \emptyset$ inside $\mathbb{N}^*$ for every n , and again $N_z \subseteq N_y$ is disjoint from U_x .

Thus no point of $\mathbb{N}^* \setminus U_x$ lies in the interior of $\overline{U_x}$. Since U_x is open, $U_x = \text{int } \overline{U_x}$, so U_x is regular open.

Next, U_x is not compact. Indeed, it is covered by the increasing open cover $\{N_{b_{x,n}} : n \geq 1\}$. No finite subfamily covers U_x , because $N_{b_{x,n+1}} \not\subseteq N_{b_{x,n}}$: the infinite set $A_{x,n} = b_{x,n+1} \setminus b_{x,n}$ yields nonprincipal ultrafilters in $N_{b_{x,n+1}} \setminus N_{b_{x,n}}$. Therefore U_x is noncompact.

Finally, if $x \neq y$, then $A_x \cap A_y$ is finite. Hence $b_{x,n} \cap b_{y,m}$ is finite for all n, m , so

$$N_{b_{x,n}} \cap N_{b_{y,m}} = \emptyset$$

inside $\mathbb{N}^*$. It follows that $U_x \cap U_y = \emptyset$, which is compact. Therefore $\{U_x : x \in 2^\omega\}$ is an almost disjoint family on $\mathbb{N}^*$ of cardinality $2^{\aleph_0}$. $\square$

Scope and Limitations

This is a partial answer to Question 4.3. It proves the ZFC lower bound $2^{\aleph_0}$. It does not prove in ZFC that there is such a family of cardinality $2^{\aleph_1}$, nor of cardinality $2^{2^{\aleph_0}}$. In models satisfying $2^{\aleph_0} = 2^{\aleph_1}$, the theorem reaches the first cardinal target in Question 4.3, but the general case where $2^{\aleph_0} < 2^{\aleph_1}$ remains outside this packet.

Novelty and Search Notes

The local run indexes were searched for arXiv id 2606.02040, the title, and core phrases around generalized almost disjoint families on $\mathbb{N}^*$. External web search was performed for the exact source question and close phrases including “ $\text{Ext}^2(c_0(\omega_1), c_0)$ ”, “injective dimension c_0 ”, and “almost disjoint family on $N^* 2^{\aleph_1}$ ”. The search surfaced the source paper itself but no separate later answer. The construction above is elementary and should be reviewed as a direct ZFC partial result rather than as a claimed literature resolution of Question 4.3.

Human Review Recommendation

The key point for review is the regular-open verification. In particular, check the thinning argument for a basic neighborhood N_y of an ultrafilter outside U_x . Once regularity is confirmed, noncompactness and pairwise compact intersections are immediate.

References

- [1] C. Lambie-Hanson and D. Schritterser, *Generalized almost disjoint families and injective Banach spaces*, arXiv:2606.02040, 2026. <https://arxiv.org/abs/2606.02040>